First Last, Ph.D.

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Experience

Senior Data Scientist *June 2022 - Present*
company name

- Utilized YOLO v5 to develop an object-detection model in PyTorch that achieved a 80% mAP evaluation on a in-house thermal-image vehicle dataset for a R&D project.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Enhanced the scalability of models initially prototyped by our team of Analysts and Data Scientists for training on our internal high-performance computing (HPC) cluster. My role involved refactoring the existing code to enable efficient execution across multiple nodes and GPUs.
- Developed interactive dashboards using Streamlit to give non-technical staff the abilty to interact with the capabilities designed by our team of Data Scientists.
- Mentored a summer intern in the development and execution of models on a HPC cluster, involving multi-node, multi-GPU setups and SLURM resource management.
- Developed a Django web interface that enabled non-technical staff to interact with the company's Neo4j graph database, allowing them to query and update data without requiring knowledge of the query syntax.

Technical Skills

Programming Languages: Python, SQL, Bash, Cypher, Lua, LaTeX

Libraries: PyTorch, Scikit-Learn, Pandas, Scipy, Sqlalchemy, Matplotlib, Plotly, Streamlit, Django

Developer Tools: Git, AWS, Unix/Linux, Slurm, Vim, Tmux, Neo4j

Education

University 1 - Ph.D. *June 2022*
Research Area: Probability (Stochastic Processes and Partial Differential Equations)
Advisor: First Last

University 2 - B.S./M.S. *December 2015 / January 2018*
Major: Applied Mathematics

Publications

1: Q1 Probability journal

2: Q2 Probability journal